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PAPER

Collective dynamics of confined rigid spheres and deformable drops†

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The evolution of linear arrays of rigid spheres and deformable drops in a Poiseuille flow between parallel walls is investigated to determine the effect of particle deformation on the collective dynamics in confined particulate flows. We find that linear arrays of rigid spheres aligned in the flow direction exhibit a particle-pairing instability and are unstable to lateral perturbations. Linear arrays of deformable drops also undergo the pairing instability but also exhibit additional dynamical features, including formation of transient triplets, cascades of pair-switching events, and the eventual formation of pairs with equal interparticle spacing. Moreover, particle deformation stabilizes drop arrays to lateral perturbations. These pairing and alignment phenomena are qualitatively explained in terms of hydrodynamic far-field dipole interactions that are insensitive to particle deformation and quadrupole interactions that are deformation induced. We suggest that quadrupole interactions may underlie the spontaneous formation of droplet strings in confined emulsions under shear [*Phys. Rev. Lett.*, 2001, **86**, 1023.].

1. Introduction

Microfluidic devices frequently rely on precise manipulation of arrays of hydrodynamically coupled particles (e.g., drops, vesicles, or biological cells).^{1–3} Such devices can be used, for example, in high-throughput biological testing,⁴ in microfabrication,^{5,6} and as microreactors.⁷ Development of sophisticated microfluidic devices requires precise control of confined multiphase flows. A detailed understanding of subtle hydrodynamic phenomena that occur in such systems is therefore essential.

In many cases particles in microfluidic flows organize themselves in ordered arrays^{8–13} (sometimes called microfluidic crystals). The simplest form of such a crystal is a linear array of regularly spaced particles moving through a channel. Hydrodynamic interactions between the particles, and between the particles and confining walls, influence the dynamics of the array, often leading to a complex collective behavior.^{9,11,13–15}

Particle arrays in cylindrical tubes and quasi-1D narrow channels^{16–19} exhibit exponential decay of interparticle hydrodynamic interactions.¹⁷ It follows that the collective behavior is controlled by short-range interactions between nearest neighbor particles.^{19–23} Particles in confined quasi-2D flows (i.e., channels with Hele-Shaw geometry) have the potential for more complex collective behavior because they have unscreened long-range hydrodynamic interactions. We focus on the collective dynamics of such systems.

Recent investigations of strongly confined particulate flows in parallel-wall channels identified the key role of a particle-scale dipolar backflow pattern.^{24–26} This backflow causes local dynamic effects such as enhanced relative particle motion,^{25,26} instabilities and particle pairing in flow-driven linear arrays,⁹ and particle realignment in a square particle lattice^{11–13} (which has been observed to form spontaneously).^{12,27} Moreover, the dipolar hydrodynamic interactions lead to macroscopic phenomena, such as propagation of displacement waves in linear arrays (“microfluidic phonons”)^{9,11,14} and the reduced collective friction coefficient in 2D arrays^{28,29} that results from a coupling of the collective particle motion and the macroscopic fluid flow.³⁰ It was shown that such coupling causes fingering instabilities in finite-size arrays.^{11,13} Finally, the dipolar backflow is involved in the complex collective dynamics of self-propelled particles.^{31,32}

Hydrodynamic interactions in confined particulate flows are also affected by the particle shape and deformability. However, most of numerical simulations and theoretical analyses dealt with systems of rigid spherical particles, even though particles (e.g., drops, vesicles) often deform in microfluidic channels. Some recent numerical studies of confined flows of deformable particles include axisymmetric boundary-integral simulation of single-file motion of red blood cells in a capillary,³³ and 3D

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boundary-integral simulations of individual drops in a parallel-wall channel.^{34,35} Only recently, with the development of new algorithms and increased computing power, have simulations of large numbers of confined drops or cells been conducted.^{36,37} None of these investigations, however, clearly identifies the hydrodynamic mechanisms through which particle deformation affects the collective evolution in strongly confined systems.

Our focus is on the influence of particle deformation on the collective dynamics of linear particle arrays confined between parallel plates under creeping-flow conditions. We consider arrays of two types of particles: rigid spheres and deformable drops. We study the effects of drop deformability, as characterized by an appropriately defined capillary number, and contrast the behavior of drops with that of rigid spheres. New collective phenomena are described, and the underlying hydrodynamic mechanisms associated with drop deformation are identified.

We consider the dynamics of individual particles and particle pairs in Poiseuille flow in Section 2, and the collective dynamics of finite-size linear particle arrays in Section 3. Section 4 presents our theoretical analysis, where we argue that the key qualitative differences between the behavior of rigid-spheres and deformable drops result from the *quadrupolar* Hele-Shaw flow associated with drop deformation. We also propose a pairwise superposition approximation for the collective dynamics of arrays of drops and use it to probe the long-time collective evolution. Finally, in Section 5, the influence of array misalignment on the collective evolution is investigated. Concluding remarks are given in Section 6.

2. Spherical particles and deformable drops in a parallel-wall channel

2.1. The system

We consider the collective dynamics of finite-length linear arrays of rigid spheres or deformable drops in Poiseuille flow between two parallel walls. The radius of the particles (undeformed drops) is a and the wall separation is $2W$ (Fig. 1). All the particles in an array are identical. The position of the center of mass of particle i is denoted by $\mathbf{R}_i = (X_i, Y_i, Z_i)$, and the migration velocity by $\mathbf{U}_i = (U_i^x, U_i^y, U_i^z)$.

As illustrated in Fig. 1, the system is driven by the incident Poiseuille flow

$$\mathbf{v}_\infty = V_\infty \left(1 - \frac{z}{W}\right) \left(1 + \frac{z}{W}\right) \hat{\mathbf{e}}_x, \quad (1)$$

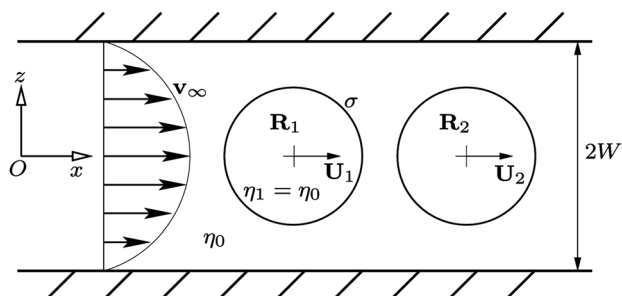


Fig. 1 Schematic of drops in Poiseuille flow between parallel walls.

where $\hat{\mathbf{e}}_x$ is the unit vector in the x direction, V_∞ is the flow amplitude, and $z = \pm W$ are the wall positions. Creeping-flow conditions are assumed in our calculations.

We focus on configurations where all particles are in the midplane of the channel. In such configurations the relative particle motion stems entirely from interparticle hydrodynamic interactions rather than from differences of the imposed velocity field at the particle positions. Particle arrangements in the mid-plane of a channel often occur in microfluidic systems, and moreover, deformable drops naturally migrate towards the channel center^{38–40} (see Video 1 in the ESI†).

To determine the effect of drop deformability on the collective dynamics of linear arrays in parallel-wall channels, we compare the evolution of linear arrays of rigid spheres and drops under similar confinement conditions. All simulations are presented for a single confinement ratio

$$W/a = 1.2. \quad (2)$$

The magnitude of drop deformation is characterized by the capillary number

$$\text{Ca} = \tau_\sigma / \tau_0, \quad (3)$$

where τ_σ is the surface-tension-driven relaxation time of the drop,

$$\tau_\sigma = \eta_1 a / \sigma, \quad (4)$$

and τ_0 is the time scale imposed by the flow,

$$\tau_0 = a / V_\infty. \quad (5)$$

Here, σ is the interfacial tension and η_1 is the drop-phase viscosity. For simplicity, we only consider the case of drops with the same viscosity as the suspending fluid, *i.e.*, $\eta_1 = \eta_0$, as indicated Fig. 1. It is also assumed that the interfacial tension is constant, *i.e.*, there are no Marangoni stresses associated with the presence of surfactant or temperature gradients.

2.2. Numerical-simulation methods

Our numerical simulations of arrays of rigid particles have been performed using Cartesian-representation method developed by Bhattacharya *et al.*^{26,41–43} This method combines the HYDRO-MULTIPOLE algorithm⁴⁴ with the expansion of the flow field into the lateral Fourier modes in the planes parallel to the walls. In the current implementation,¹¹ we take advantage of the far-field asymptotic behavior of the Green's functions in the parallel-wall geometry^{24,30,42} and utilize sparse-matrix techniques to significantly reduce the numerical cost of the simulations. Typically, simulations of arrays of rigid spheres required a few minutes of CPU time.

For the study of deformable drops we apply the boundary-integral algorithm developed by Janssen and Anderson.^{35,45} Drop surfaces in arrays of three or more drops were discretized with $N = 2000$ triangular elements each; higher resolution meshes ($N = 8000$ elements per drop) were used for single-drop and pair computations. In the near-field regime, boundary-integrals are evaluated using the exact Green's functions for Stokes flow in a parallel-wall channel.^{24,46} In the far-field, explicit asymptotic

expressions are used in order to speed up the calculations by avoiding the costly evaluation of Fourier–Bessel integrals.^{24,42} Nevertheless, very long computation times (up to months) were required to simulate drop arrays to reveal their collective dynamics. The large computational cost for the boundary-integral calculations stems from two factors. The first is the high cost of evaluating the Green's functions, and the second is the need to resolve the capillary time on a lengthscale set by the size of the boundary elements (*i.e.*, $\Delta t < \tau_\sigma/\sqrt{N}$) to assure numerical stability. The very high computational cost of the boundary integral simulations motivates the pairwise superposition approximation described in Section 4.2.

2.3. Pair dynamics and drop deformation effects

Before investigating the collective dynamics of multi-particle flow-driven arrays, we first consider simple one- and two-body systems. In Section 2.3.1 we investigate flow-induced drop deformation, and in Section 2.3.2 we compare the evolution of pairs of drops and pairs of rigid spheres to reveal the effect of deformation on hydrodynamic interactions between particles. It is assumed that the particle pairs are aligned with the imposed velocity (1). Misaligned pairs are considered in Section 5.

2.3.1. Drop deformation. Flow-induced drop deformation in a parallel-wall channel is illustrated in Fig. 2. The shape of an isolated drop is depicted at several capillary numbers in Fig. 2(a); the effect of pair hydrodynamic interactions on the drop shape is shown in Fig. 2(b). Due to the asymmetric form of the imposed flow (1), a deformed drop lacks fore–aft symmetry, which has important consequences for drop dynamics.

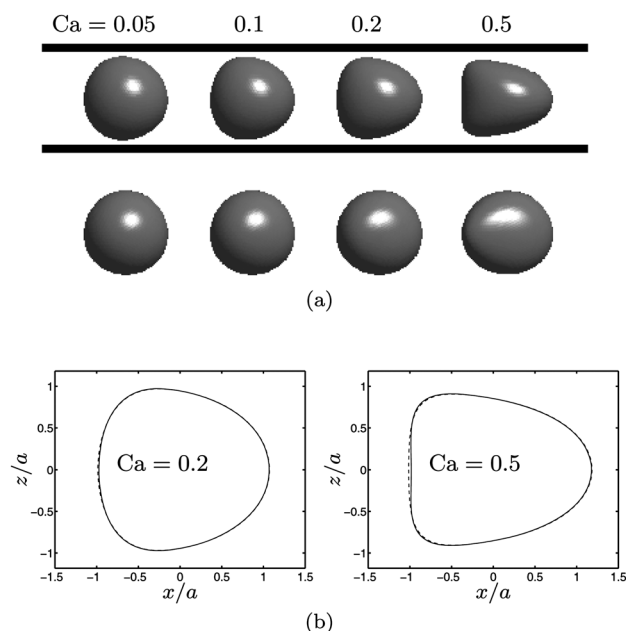


Fig. 2 (a) Steady shapes (side and top views) of isolated drops in Poiseuille flow in a parallel-wall channel. Capillary numbers as indicated; wall location shown. (b) Comparison of the shape of an isolated drop (dashed line) with the shape of the leading drop in a stationary pair (solid line), capillary numbers as indicated.

The results presented in Fig. 2(b) indicate that the effect of hydrodynamic interactions between drops on the drop shape is quite small, even at small drop separations. Thus, the stationary shape of an isolated drop is relevant for multiparticle dynamics, as discussed below.

2.3.2. Hydrodynamic interactions of particle pairs. Owing to the flow-reversal symmetry of Stokes flow,⁴⁷ hydrodynamic interactions between two rigid spheres in the midplane of the channel cannot produce relative particle motion. Hydrodynamic interactions only affect the velocity of the center of mass of the particle pair. By contrast, two drops have a nonzero relative velocity because drop deformation (*cf.* Fig. 2) removes the fore–aft symmetry (see Video 2 in the ESI†).

These effects are illustrated in Fig. 3 and 4 for rigid spherical particles and deformable drops, respectively. Fig. 3 and 4(a) show the center-of-mass velocity,

$$U_{\text{CM}} = \frac{1}{2}(U_1^x + U_2^x), \quad (6)$$

of a particle pair aligned in the flow direction *versus* particle separation $X_{12} = X_2 - X_1$. The relative velocity,

$$U_{12} = U_2^x - U_1^x, \quad (7)$$

of two deformable drops at different capillary numbers is depicted in Fig. 4(b).

The results shown in Fig. 3 and 4(a) indicate that the center-of-mass velocity U_{CM} is larger for drops than for rigid spheres. We also find that for drops the velocity U_{CM} increases with capillary number, due to the larger gaps that form between the drop interface and the walls with increasing deformation (*cf.* Fig. 2) and the fact that the flattened drop samples a higher velocity portion of the imposed flow (1).

The center-of-mass velocity U_{CM} decreases with the decreasing interparticle distance X_{12} for rigid particles and for drops, according to the results shown in Fig. 3 and 4(a), consistent with our earlier calculations for rigid spheres with different confinement ratios.⁴³ This decrease stems from the interparticle hydrodynamic interactions that cause a reduction of the hydrodynamic

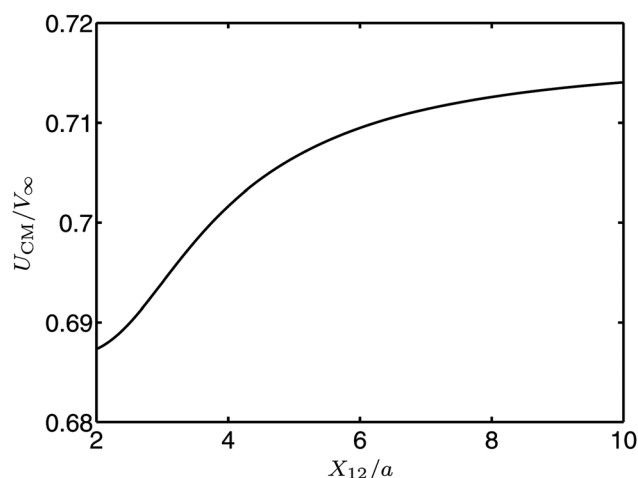


Fig. 3 Center-of-mass velocity U_{CM} for a pair of rigid spheres aligned with the imposed velocity (1) *versus* particle separation X_{12} .

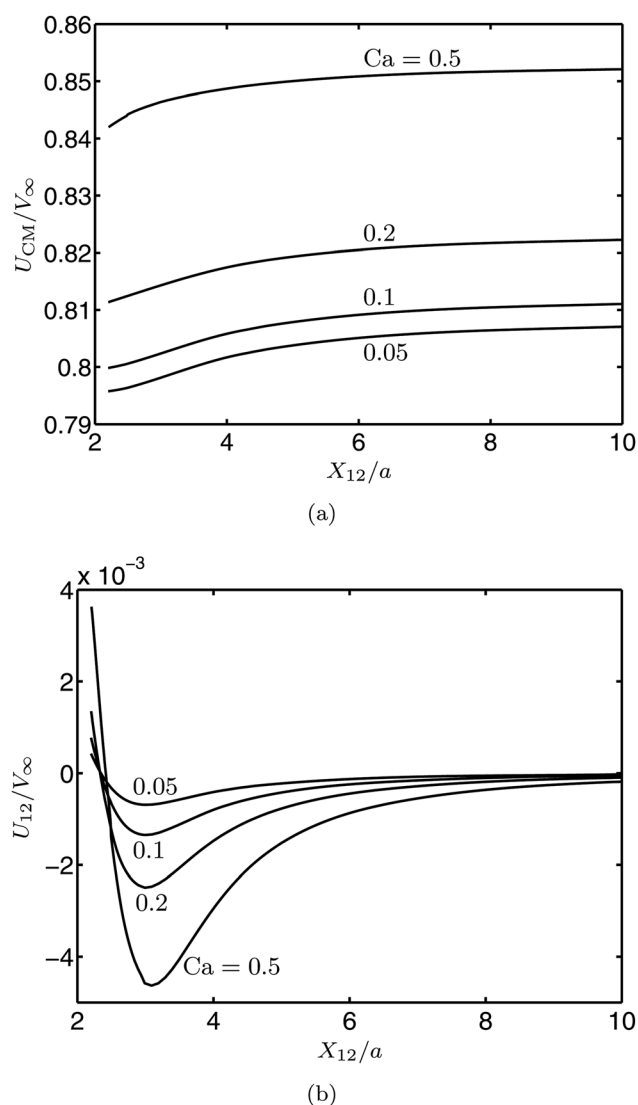


Fig. 4 (a) Center-of-mass velocity U_{CM} and (b) relative velocity U_{12} for a pair of drops aligned with the imposed velocity (1) versus drop separation X_{12} ; capillary numbers as indicated.

drag exerted by the external flow on the pair. At large separations X_{12} , the velocity U_{CM} tends to the single-particle value, $U_{CM} \rightarrow U$.

For rigid spheres, the limiting value $U/V_\infty = 0.717$ is lower than the estimate $U/V_\infty = 0.769$ provided by Faxén's law:⁴⁷

$$U = \left[\mathbf{v} + \frac{a^2}{6} \nabla^2 \mathbf{v} \right]_{z=0} \quad (8)$$

for a particle in an unconfined flow.[§] A similar result holds for deformable drops, where the limiting values U/V_∞ for all capillary numbers are lower than the value $U/V_\infty = 0.861$ obtained from the generalized Faxén's formula:⁴⁸

$$U = \left[\mathbf{v} + \frac{3\eta_1}{2\eta_0 + 3\eta_1} \frac{a^2}{6} \nabla^2 \mathbf{v} \right]_{z=0} \quad (9)$$

[§] The value $U/V_\infty = 0.769$ is obtained from formula (8) using velocity field (1) with confinement ratio (2) but otherwise ignoring wall effects.

for a spherical drop with the viscosity ratio $\eta_1/\eta_0 = 1$. The observed deviations from the Faxén-law predictions (8) and (9) result from hydrodynamic interactions of particles with channel walls and from drop deformation.

The relative motion of deformable drops stems from the asymmetry of their deformed shape. Thus, the magnitude of the relative velocity U_{12} increases with capillary number, as seen in Fig. 4(b). The results show that hydrodynamic interactions cause drops to attract each other at large separations and repel each other at small separations, the relative velocity changing sign at a critical separation $X_0 \approx 2.3a$. Accordingly, drops tend to a stable separation $X_{12} = X_0$ at long times, as illustrated in Fig. 5. The rate at which the stationary configuration is achieved increases with deformation, consistent with the relative velocity (Fig. 4(b)). However, the stationary separation X_0 is nearly independent of the capillary number.

The distinction between deformable drops which tend to a stable separation and rigid spheres which do not, gives rise to interesting qualitative differences between their collective dynamics in linear arrays, as shown in Sections 3 and 4.

3. Pairing instability in linear arrays of rigid spheres and deformable drops

Here we describe the collective behavior of linear arrays of rigid spheres and deformable drops. The arrays are aligned with the direction of the imposed velocity (1), and at the beginning of the evolution the particles are equally spaced. We assume that drops are initially spherical, but this assumption has a negligible impact on the array dynamics for $t > \tau_\sigma$ [*i.e.*, times larger than the drop-relaxation time (4)].

Examples of the collective behavior of particle arrays are shown in Fig. 6 and 8 for rigid spheres and in Fig. 7, 9, and 10 for deformable drops. Note that trajectories (Fig. 8–10) are depicted in the reference frame of the trailing particle in the array. While we only present results for ten-particle arrays, we expect that the qualitative features that they illustrate should hold for all finite-length arrays with more than two particles.

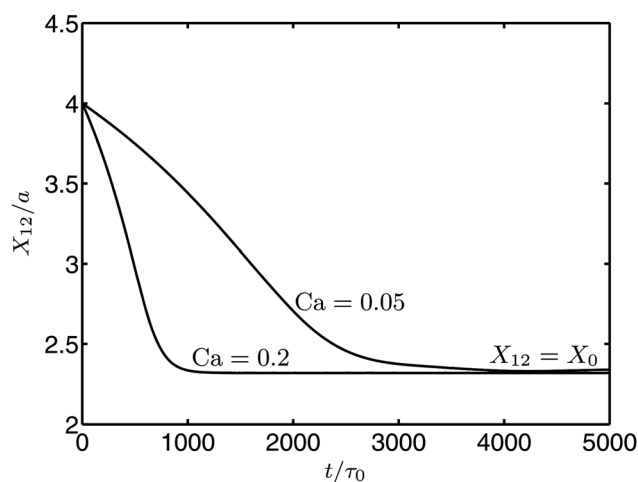


Fig. 5 Time evolution of the drop separation X_{12} for a pair of drops aligned with the imposed velocity (1); capillary numbers as indicated, time normalized by the flow time scale (5).

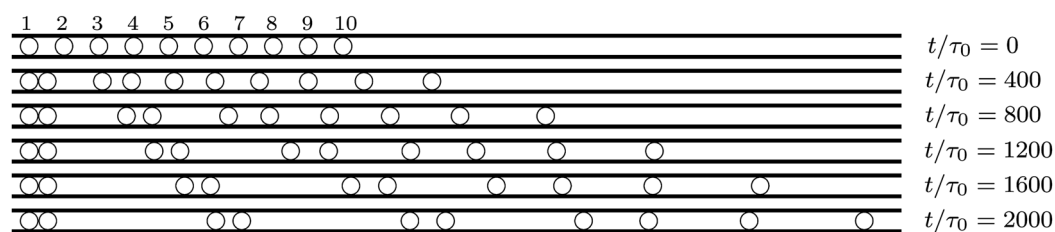


Fig. 6 Evolution of a linear array of rigid spheres aligned in the flow direction for the initial particle separation $\Delta X = 4a$; time as labeled. See Video 3 in the ESI†.

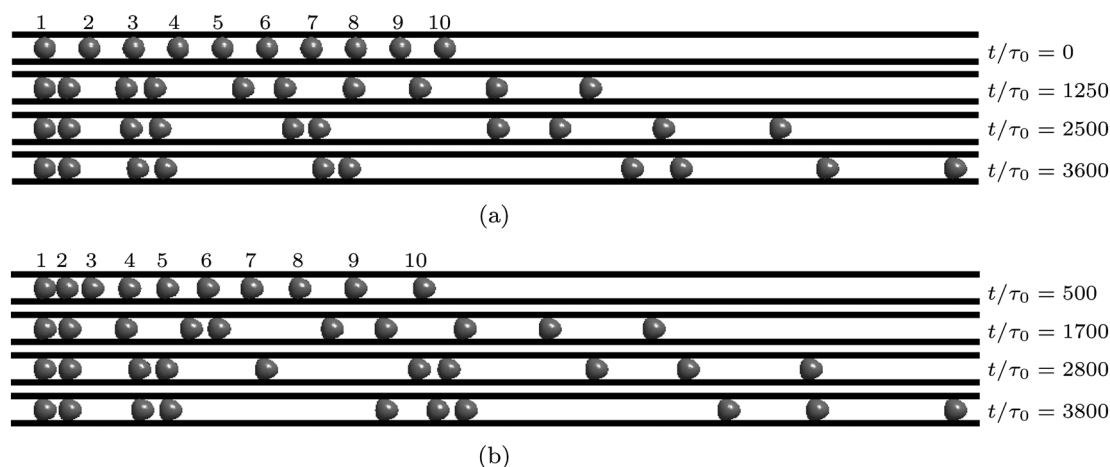


Fig. 7 Evolution of linear arrays of deformable drops aligned in the flow direction for the initial drop separation (a) $\Delta X = 4a$ and (b) $\Delta X = 3a$; capillary number $Ca = 0.2$, and time as labeled. See Videos 4 and 5 in the ESI†.

3.1. Pairing instability

Fig. 6 shows several snapshots of the particle configuration at different times for an array of rigid spheres with initial separation $\Delta X = 4a$; the corresponding particle trajectories are depicted in Fig. 8(a). The results indicate that the system undergoes a pairing instability that starts at the back of the array and propagates forward. In the initial phase of the evolution, particle 1 catches up with particle (2) forming a pair, which moves slower than the other particles, consistent with the results depicted in Fig. 3. Thus, the pair falls behind the rest of the array, and the pairing process is then repeated for particles 3 and 4, 5 and 6, and so on, leading to a decomposition of the array into a succession of particle pairs.

We find that the pairing instability is a generic feature of the array evolution in parallel-wall channels. It occurs for arrays with different initial particle separations [e.g., Fig. 8(a) for $\Delta X = 4a$, Fig. 8(b) for $\Delta X = 3a$], different confinement ratios W/a (not shown), and for arrays of deformable drops (Fig. 7, 9 and 10). The pairing instability is also evident in the numerical results of McWhirter *et al.*²³ for red blood cells in pressure-driven flow within a cylindrical tube (e.g., Fig. 5 in their paper).

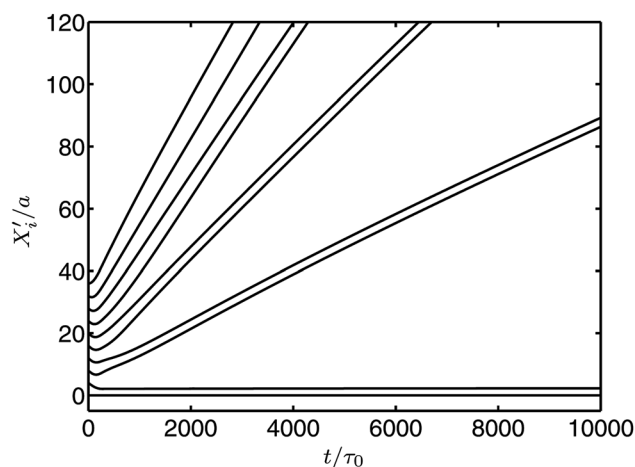
The tendency for pairs of deformable drops to achieve a stable separation at long times (*cf.* Fig. 5) is reflected in the shape of the trajectories depicted in Fig. 9 and 10. After formation of a pair, the distance between drops gradually decreases, until the stationary separation X_0 is achieved. The tendency towards a unique stable separation is more evident in Fig. 11, which

shows long-time evolution of drop separation in drop pairs that have formed as a result of array evolution.

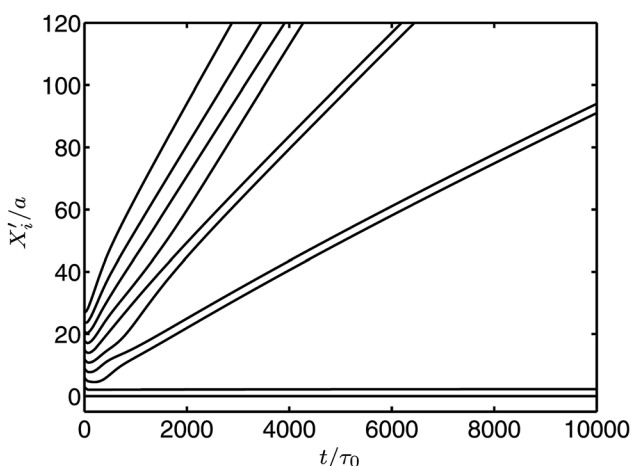
The time variation of the velocity of drop pairs, which according to Fig. 4(a) depends on the relative drop position, is consistent with the evolution of drop separation. The long-time velocities vary slightly from pair to pair because of the hydrodynamic interactions between the pairs. However, this variation is significantly smaller than the variation for rigid spheres, as seen in Fig. 8. In the rigid-particle case the ultimate separation between spheres in a pair is variable and determined by the initial conditions, *i.e.*, the location of the pair within the array and the initial separation between the particles, ΔX .

3.2. Pair-switching cascade

For sufficiently large capillary numbers (*i.e.*, deformability) and close initial spacing, arrays of deformable drops undergo a cascade of pair-switching events at moderate times that are associated with the pairing instability; an example is shown in Fig. 7(b). Here, a temporary triplet (drops 1–3) forms (*cf.* the configuration at $t/\tau_0 = 500$), the lead drop of the triplet (drop 3) separates and advances to the temporary pair 4–5 ($t/\tau_0 = 1700$) triggering a pair switching event that forms the pair 3–4 with drop 5 advancing ($t/\tau_0 = 2800$). The process repeats with subsequent pairs, and a cascade of pair-switching events propagates to the front of the array. Pair switching cascades are also evident from the trajectories depicted in Fig. 9(b) and in Fig. 10. For smaller capillary numbers and larger initial separations, this



(a)



(b)

Fig. 8 Time evolution of particle centers relative to the position of the trailing particle, $X'_i = X_i - X_1$, for a linear array of rigid spheres aligned in the flow direction; initial particle separations (a) $\Delta X = 4a$ and (b) $\Delta X = 3a$.

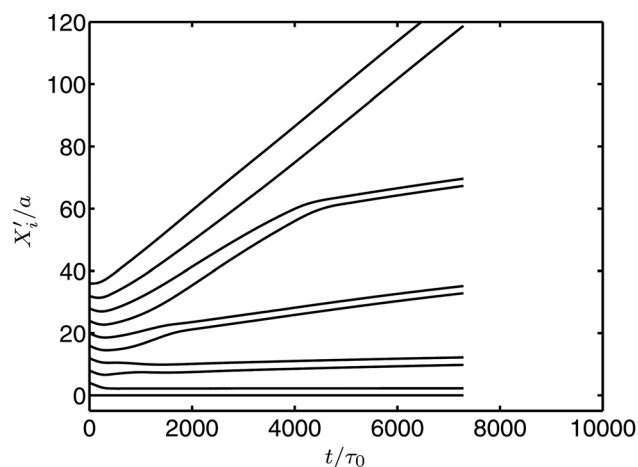
cascade of pair-switching does not occur [e.g., Fig. 7(a) and 9(a)]. Under these conditions, the pairing instability for drops resembles the behavior of rigid spheres at short times, but eventually, the drops in each pair tend to their intrinsic stable separation.

4. Hele-Shaw multipoles and superposition models

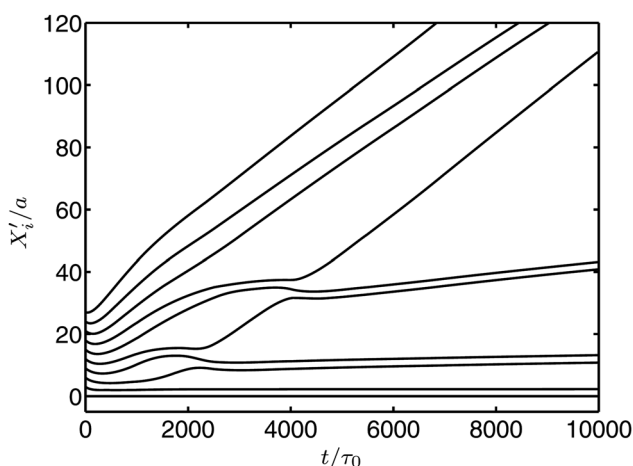
In Section 4.1 we discuss the hydrodynamic mechanisms responsible for the collective phenomena described above, and in Section 4.2 we develop a pairwise-superposition approximation.

4.1. Far-field hydrodynamic interactions

4.1.1. Hele-Shaw far-field flow. Interparticle hydrodynamic interactions in Poiseuille flow between two parallel walls differ significantly from hydrodynamic interactions in free space. In strongly confined flows, the collective particle dynamics are qualitatively modified because of the distinctive far-field behavior of the flow scattered by the particles.⁴⁹ In a confined



(a)



(b)

Fig. 9 Same as Fig. 8, except for drops with capillary number $Ca = 0.2$.

flow, the scattered flow in the far-field regime tends to the quasi-two-dimensional Hele-Shaw form

$$\mathbf{v}' = -\frac{1}{2}\eta_0^{-1}(W-z)(W+z)\nabla p'(x,y), \quad (10)$$

where the asymptotic two-dimensional pressure distribution depends only on the lateral position $\boldsymbol{\rho} = x\hat{\mathbf{e}}_x + y\hat{\mathbf{e}}_y$.

For sufficiently large interparticle distances (greater than several wall separations) interparticle hydrodynamic interactions in a parallel-wall channel occur only through such Hele-Shaw fields, most important of which are the dipolar and quadrupolar contributions associated with the pressure distributions

$$p_d(\boldsymbol{\rho}) = -\frac{x}{\rho^2} \quad (11)$$

and

$$p_q(\boldsymbol{\rho}) = -\frac{x^2 - y^2}{\rho^4}, \quad (12)$$

where $\rho = |\boldsymbol{\rho}|$. The orientation of the fields (11) and (12) follows from the assumption that the imposed velocity (1) is in the x direction. The role of the dipolar and quadrupolar fields (11) and (12) in multiparticle systems is discussed below.

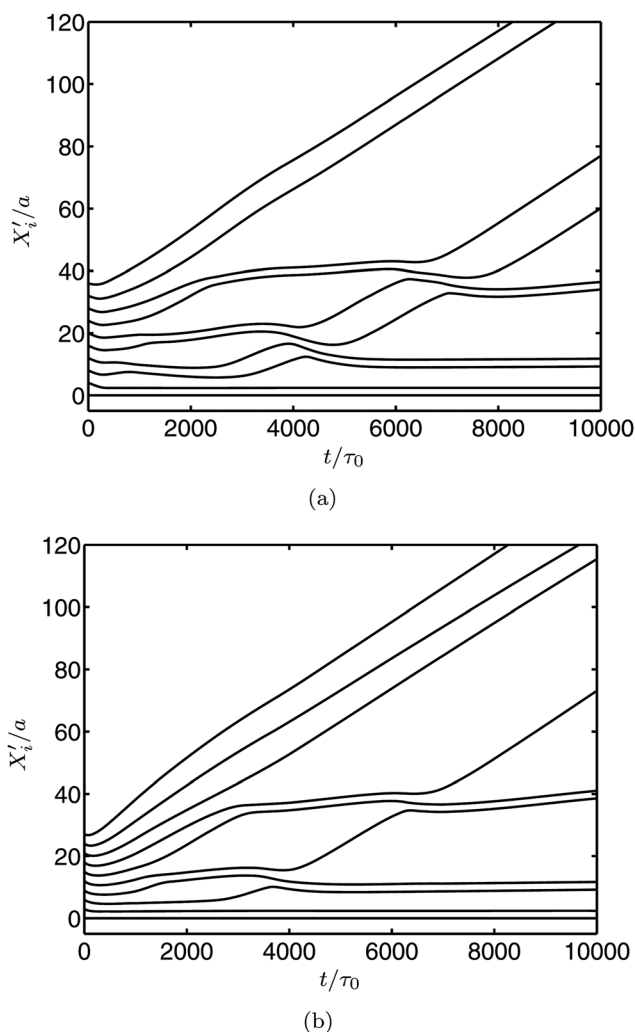


Fig. 10 Same as Fig. 8, except for drops with capillary number $Ca = 0.5$.

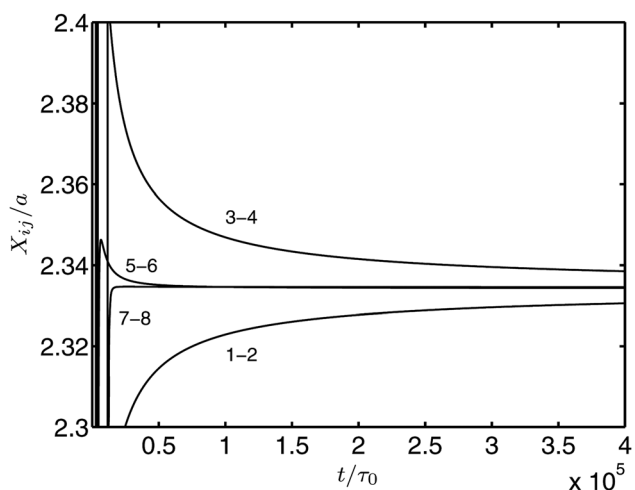


Fig. 11 Long time evolution of drop separation $X_{ij} = X_j - X_i$ in particle pairs (as labeled) in a linear drop array with $Ca = 0.2$ and initial separation $\Delta X = 3a$. Simulated using pairwise-superposition approximation described in Section 4.2.

4.1.2. Dipolar and quadrupolar contributions. For an isolated spherical particle in an imposed Poiseuille flow (1), the far-field scattered flow (10) involves only the dipole (11)

$$p'(\rho) = Dp_d(\rho), \quad (13)$$

where D is the dipole moment which depends on the particle size and wall separation. The quadrupolar and higher-order contributions vanish by symmetry. For a deformable drop, the imposed flow (1) couples to the non-spherical drop shape, which, to the leading order in the capillary number, yields

$$p'(\rho) = Dp_d(\rho) + CaQp_q(\rho), \quad (14)$$

where CaQ is the quadrupole moment.

In multiparticle systems, the leading-order hydrodynamic interactions inherit the multipolar structure of the scattered flows (13) and (14). Thus, for the velocity of a rigid sphere (i) in an array we obtain

$$U_i = \mu_0 V_\infty \hat{e}_x + \mu_0 \sum_{j \neq i} v_d(\rho_{ij}), \quad (15)$$

where $\mu_0 = U/V_\infty$ is the mobility of an isolated particle in the imposed parabolic flow (1), $\rho_{ij} = \rho_j - \rho_i$ is the relative particle position (with $\rho_k = X_k \hat{e}_x + Y_k \hat{e}_y$), and

$$v_d(\rho) = -\frac{1}{2} \eta_0^{-1} W^2 D \nabla p_d(\rho) \quad (16)$$

is the magnitude of the Hele-Shaw dipolar flow in the midplane $z = 0$. For deformable drops we have an analogous expression,

$$U_i = \mu_0 V_\infty \hat{e}_x + \mu_0 \sum_{j \neq i} v_d(\rho_{ij}) + \mu'_0 \sum_{j \neq i} v_q(\rho_{ij}), \quad (17)$$

where

$$v_q(\rho) = -\frac{1}{2} \eta_0^{-1} W^2 Ca Q \nabla p_q(\rho) \quad (18)$$

is the quadrupolar Hele-Shaw flow, and μ'_0 is the drop mobility that combines the mobility of an isolated drop, μ_0 , defined above, plus an additional contribution resulting from the interaction of the dipolar flow v_d produced by drop j with the deformed shape of drop i .

The dipolar and quadrupolar flow fields (16) and (18) are schematically depicted in Fig. 12 and 13. Because of the distinct parity of these flow fields, the dipolar and quadrupolar interactions affect, respectively, only the center-of-mass velocity U_{CM} and relative velocity U_{12} of an isolated particle pair. This property combined with relations (11) and (12) implies that

$$U_{CM} - \mu_0 V_\infty \sim \rho^{-2} \quad (19)$$

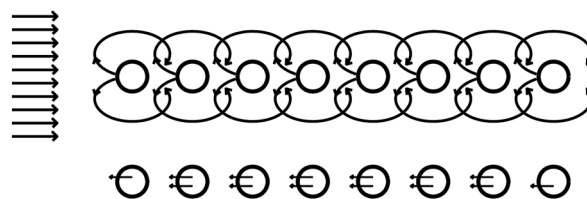


Fig. 12 Schematic representation of the pairing mechanism due to dipolar interactions. Neighboring particles slow each other down.

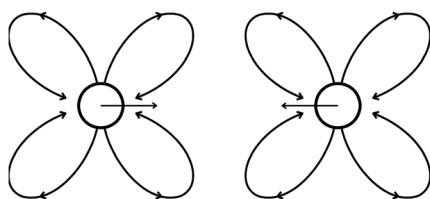


Fig. 13 Quadrupole interactions between drops lead to hydrodynamic attraction between them.

and

$$U_{12} \sim Ca\rho^{-3}, \quad (20)$$

as illustrated in Fig. 14 for a drop pair oriented in the flow direction.

4.1.3. Qualitative analysis of pairing instability

4.1.3.1. Dipolar contribution. Using arguments similar to those proposed by Beatus *et al.*,⁹ we find that the dipolar contribution (15) to the hydrodynamic interactions between particles captures most of the essential features of the pairing instability discussed in Section 3.1; the mechanism is illustrated in Fig. 12. The schematic shows that each particle produces the Hele-Shaw dipolar flow (16), which acts on the neighboring particles, slowing them down. Since the leading and trailing particles have only a single neighbor, they move faster than the remaining particles.

The faster-moving trailing particle in the array catches up with the next one, forming a pair with a small interparticle separation. The closely bound pair moves more slowly due to the stronger dipolar interactions, and thus falls behind the rest of the array. This creates the condition for the formation of the next pair *via* the same mechanism, and the process repeats until the array has decomposed into a sequence of particle pairs. At the front of the array, the faster-moving lead particle escapes from the array, causing the next particle to become the faster-moving leader. This process repeats, leading to a monotonic increase of the pair spacing from the back to the front of the array. Fig. 6 and 8 illustrate these processes.

Dipolar interactions are, however, insufficient to explain two distinctive features of linear arrays of deformable drops: (1) the tendency for drop pairs to achieve an intrinsic stable separation and (2) the cascade of pair-switching that occurs for sufficient deformation.

4.1.3.2. Quadrupolar contribution. Quadrupole interactions (and higher-order odd multipoles) induced by drop deformation give rise to the relative motion between drops in a pair (*cf.* Fig. 4(b)). The rescaled relative velocity, plotted in Fig. 14(b), indicates that the $O(\rho^{-3})$ quadrupolar contribution (18) dominates the relative motion of widely separated drops. The schematic in Fig. 13 shows how the quadrupolar Hele-Shaw field (18) gives rise to hydrodynamic attraction between drops.

At smaller separations, higher-order terms reduce the hydrodynamic attraction and, at a critical separation $X_{12} = X_0$, cause the sign change of the relative drop velocity, as seen in Fig. 4(b) and 14(b). As a result of the far-field quadrupolar attraction and

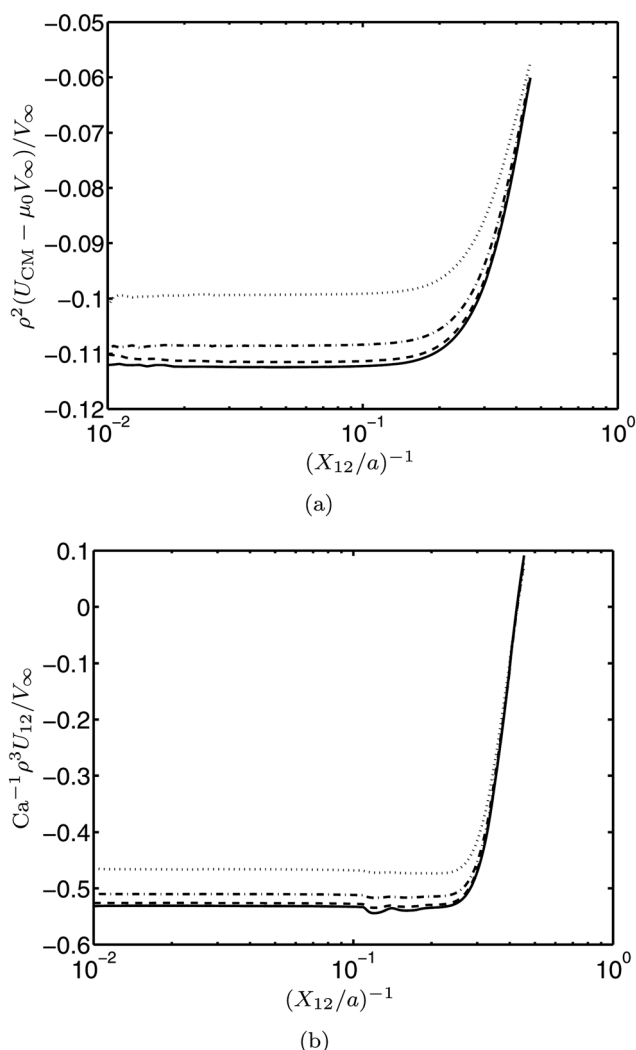


Fig. 14 Rescaled (a) center-of-mass velocity and (b) relative velocity *versus* inverse drop separation for a pair of drops with capillary number $Ca = 0.05$ (solid line), $Ca = 0.1$ (dashed), $Ca = 0.2$ (dash-dotted), and $Ca = 0.5$ (dotted).

the short-range hydrodynamic repulsion, all pairs of drops produced by the pairing instability ultimately achieve the same stable separation, $X_{12} = X_0$.

Long-range attraction and short-range repulsion also help to explain the formation of a temporary triplet at the trailing end of drop arrays. The triplet formed from drops 1–3 in Fig. 7(b) occurs when the attraction between drops 1 and 2 is small because $X_{12} \approx X_0$, while the attraction between drops 2 and 3 is sufficiently strong to draw the three drops together.

4.2. Pairwise superposition approximation

In this section we introduce a pairwise superposition approximation that incorporates the near- and far-field effects discussed above. Accordingly, the particle velocities are given by

$$\mathbf{U}_i = \mu_0 V_\infty \hat{\mathbf{e}}_x + \sum_{j \neq i} \mathbf{u}_2(\boldsymbol{\rho}_{ij}), \quad (21)$$

where the velocity

$$\mathbf{u}_2(\boldsymbol{\rho}_{12}) = \mathbf{U}_1(\boldsymbol{\rho}_{12}) - \mu_0 V_\infty \hat{\mathbf{e}}_x \quad (22)$$

is obtained from two-particle simulations. Here $\boldsymbol{\rho}_{12}$ is the relative position of particles 1 and 2, and $\mathbf{U}_1(\boldsymbol{\rho}_{12})$ is the velocity of particle 1 in an isolated pair 1–2.

Since our focus is on the dynamics of linear arrays, in what follows we consider only particles aligned in the x direction. Thus we write $\mathbf{u}_2 = u_2 \hat{\mathbf{e}}_x$, where u_2 is the magnitude of the two-particle velocity, discussed in Section 2.3.2. In our calculations we represent our pair simulation results as an expansion in inverse powers of particle separation,

$$u_2(X_{12}) = \sum_{k=2}^{k_{\max}} \frac{\alpha_k}{X_{12}^k}, \quad (23)$$

where α_k are constants. Accordingly, $k = 2$ corresponds to the dipolar contribution (16) and $k = 3$ to the quadrupolar contribution (18). The results presented in Fig. 15 were obtained with the truncation level $k_{\max} = 14$.

Note that even terms in the expansion (23) affect only the velocity of center of mass of a particle pair, U_{CM} , and the odd terms affect only the relative velocity U_{12} . Deformable drops require both odd and even terms, whereas all odd contributions vanish for rigid spheres because an isolated pair of rigid spheres does not undergo relative motion.

For rigid spheres there is no short-range pairwise hydrodynamic repulsion to keep the particles apart, thus approximate trajectories obtained from the superposition approximation may lead to particle overlap. This deficiency can be avoided by including lubrication corrections in the analysis.^{26,41,50} However, lubrication corrections are pair-additive on the friction-matrix level (rather than mobility), so including them would significantly complicate the approximation. We thus implemented the *ad hoc* additive correction

$$u_2'(X_{12}) = \exp[-A(X_{12} - d_{\text{eff}})], \quad (24)$$

which prevents particle overlap, while preserving the simple form of the superposition approximation. (A similar approach is often used in point-particle approximations.)

A comparison of numerical simulations performed using the superposition approximation with the corresponding results of

the exact numerical calculations is depicted in Fig. 15. The results for rigid spheres (*cf.* Fig. 15(a)) were obtained using correction (24) with $A = 20$ and $d_{\text{eff}} = 2.01a$; no such *ad hoc* correction was used for deformable drops. The results shown in Fig. 15 and similar calculations performed for different parameter values demonstrate that the pairwise superposition approximation reproduces fairly well the collective dynamics of linear arrays of rigid spheres and deformable drops. Given the computation cost of the boundary integral simulations, calculations based on this approximation are useful for exploring the dynamics of large systems (*e.g.*, two-dimensional drop arrays) and the evolution of linear drop arrays at very long times. An example of the latter is depicted in Fig. 11, which shows the approach to the stationary value of drop separation in each of the pairs formed from the pairing instability in an array of drops.

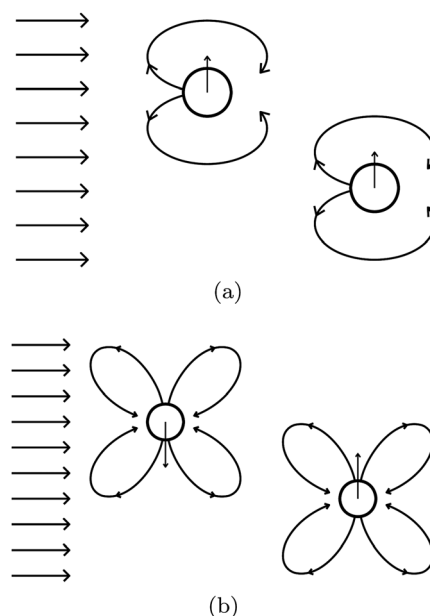


Fig. 16 Hydrodynamic interactions with lateral displacements. (a) Dipole interactions cause lateral drift of a particle pair in the y -direction. (b) Quadrupole interactions align drops with the imposed velocity (1).

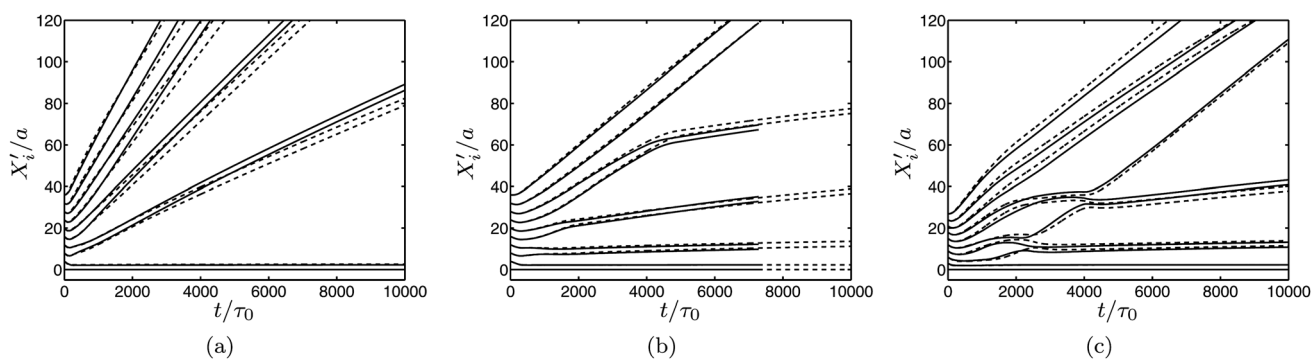


Fig. 15 Time evolution of particle centers relative to the trailing particle, $X'_i = X_i - X_1$, for linear arrays. Comparison of the pairwise-superposition approximation (21) (dashed lines) with exact numerical simulations (solid) for (a) rigid spheres, with $\Delta X = 4a$, (b) drops $\text{Ca} = 0.2$ with $\Delta X = 4a$, and (c) drops $\text{Ca} = 0.2$ with $\Delta X = 3a$.

5. Effect of lateral displacements on array dynamics

Here we consider the collective dynamics of linear arrays that are misaligned with the external velocity (1). First, we consider the dynamics of misaligned rigid-sphere and deformable-drop pairs, as illustrated in Fig. 16. Then we present results for the collective dynamics of linear arrays where one particle in the array is initially misaligned.

5.1. Lateral displacements in particle pairs

Rigid spherical particles in an isolated pair do not undergo relative motion, as discussed in Section 2.3.2; however, a pair misaligned with the imposed flow drifts laterally, in the y -direction, due to the anisotropy of the collective mobility tensor.¹³ The lateral drift of particle pairs can be intuitively explained by dipole-dipole interactions,⁹ as illustrated in Fig. 16(a).

Pairs of drops also exhibit lateral drift but, unlike pairs of rigid spheres, drops also undergo deformation-driven relative motion, ultimately resulting in their stable separation and alignment with

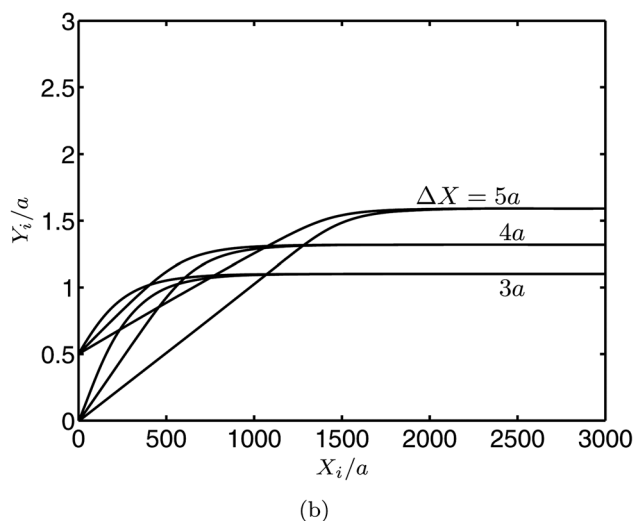
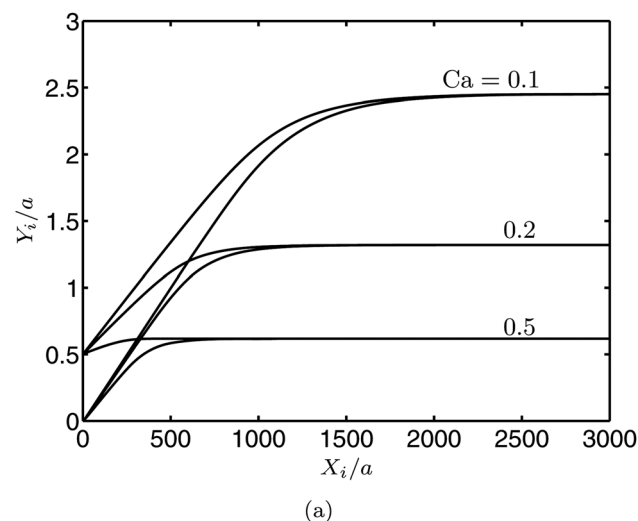


Fig. 17 The drop trajectories for two drops, with initial displacement $\Delta Y = 0.5a$, (a) $\Delta X = 4a$, Ca as indicated and (b) $Ca = 0.2$, ΔX as indicated.

respect to the imposed velocity (1). The tendency to achieve a stable separation is discussed in Section 2.3.2. The tendency for a pair of deformable drops to reorient is illustrated in Fig. 17, where we show the evolution of the lateral positions Y_1 and Y_2 for drops in isolated pairs with different capillary number and longitudinal separation ΔX . Initially, $Y_1 \neq Y_2$, causing the drops to drift laterally; however, they ultimately attain the same lateral

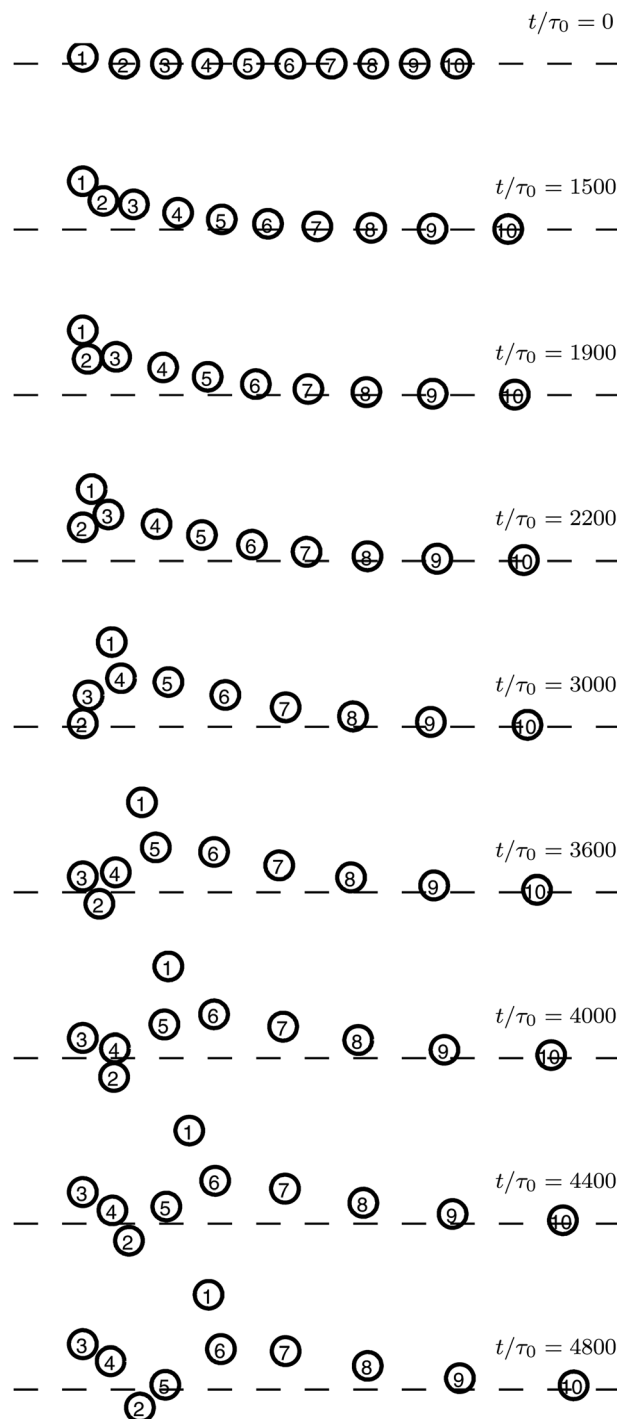


Fig. 18 Evolution of a misaligned array of rigid spheres (view in x - y plane); initial lateral particle positions are $Y_1 = 0.5a$, $Y_i = 0$ for $i = 2, \dots, 10$; initial longitudinal particle separation is $\Delta X = 3a$; time as labeled. See Video 6 in the ESI†.

position ($Y_1 = Y_2$) and stop drifting. The results show that the rate of alignment is faster for larger capillary numbers and for smaller initial particle separations, consistent with the rate at which drops attain their stable separation (*cf.* Fig. 5). The tendency for drop alignment with the flow can be qualitatively explained in terms of quadrupolar hydrodynamic interactions between the drops, as illustrated in Fig. 16(b). Note that quadrupolar and dipolar contributions have opposing effects on pair orientation, according to Fig. 16(a) and (b).

5.2. Lateral displacements in particle arrays

The effect of a lateral displacement of the trailing particle on the behavior of an array of rigid spheres (with all other particles aligned in the flow direction) is illustrated in Fig. 18; the corresponding results for an array of deformable drops are shown in Fig. 19. The collective dynamics illustrated in these figures stem from the combination of the pairing instability and the lateral drift/alignment mechanisms. The results show that arrays of rigid spheres and deformable drops exhibit qualitatively distinct collective dynamics. In the case of rigid spheres (*cf.* Fig. 18), the initial lateral perturbation at the trailing end grows, and the array decomposes into a disordered group of particles. By contrast, the tendency of deformable drops to align in the flow direction stabilizes the drop array (*cf.* Fig. 19). At first, the drops drift in the direction of the initial perturbation but the magnitude of the lateral velocities decays in time, and the array re-aligns with the imposed velocity (1). The pairing instability and pair-switching cascade are also visible in Fig. 19, much like an aligned array of drops with the same parameters (*cf.*, Fig. 9).

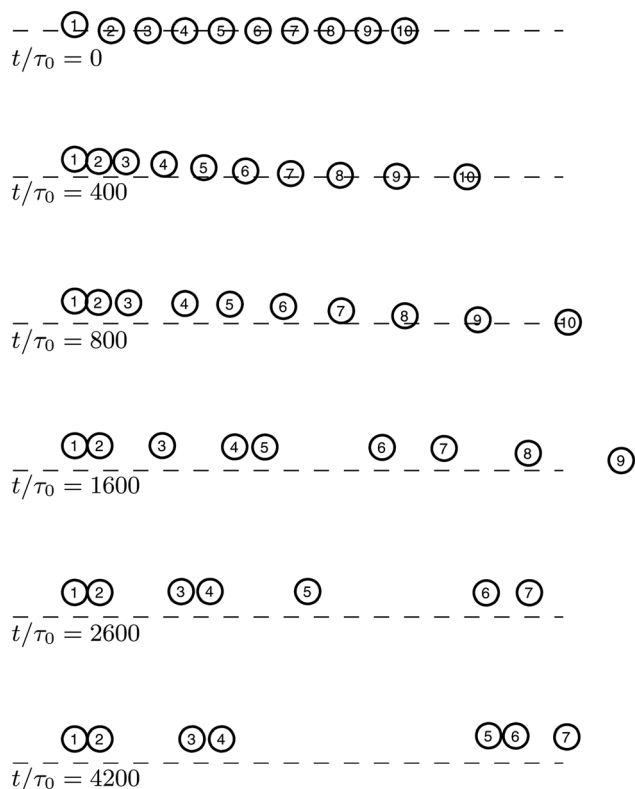


Fig. 19 Same as Fig. 18, except for drops with $Ca = 0.2$; drops represented by circles with radius a . See Video 7 in the ESI†.

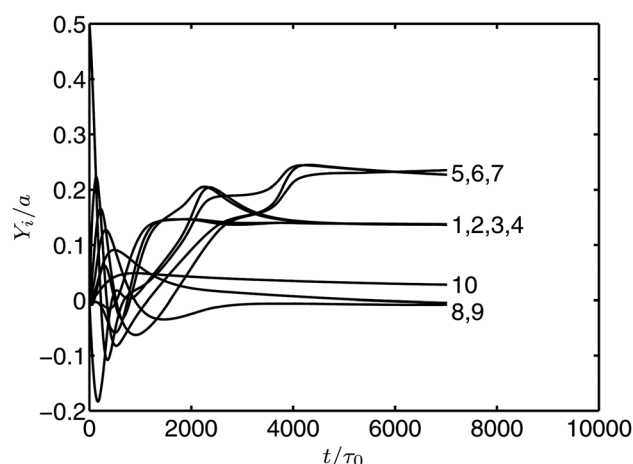


Fig. 20 Evolution of lateral positions Y_i of drops in a misaligned array, $Ca = 0.2$; initial lateral particle positions, $Y_5 = 0.5a$ and $Y_i = 0$ for $i \neq 5$; initial longitudinal particle separation is $\Delta X = 3a$.

Fig. 20 depicts the collective dynamics in a 10-drop array where drop 5 (rather than the trailing drop) is laterally perturbed. Oscillatory lateral drop displacements are observed, similar to those previously seen for arrays of rigid spheres.¹¹ Here, however, the tendency for deformable drops to align leads to the decay of these oscillations. Ultimately, smaller clusters of flow-aligned drops are produced, similar to the result seen in Fig. 19.

The stability of linear arrays of drops seen in Fig. 19 and 20 is reminiscent of the droplet strings that form in confined emulsions under shear,^{51–55} suggesting that the same quadrupolar alignment mechanism may be responsible.

6. Conclusions

The dynamics of flow-driven arrays of rigid spheres and deformable drops in parallel-wall channels were investigated and the effect of drop deformation on the collective evolution under Stokes-flow conditions was explored. Our analysis reveals the complex behavior of such arrays and identifies key hydrodynamic mechanisms responsible for the observed differences between the collective dynamics of rigid spheres and deformable particles.

We find that finite-size linear arrays aligned with the imposed velocity undergo a pairing instability that propagates forward from the back of the array and results in its decomposition into particle pairs. Drop pairs produced by the pairing instability each ultimately achieves the same stable interparticle separation, while the interparticle distance in pairs of rigid spheres has a polydisperse distribution that is sensitive to the initial conditions. For strongly deformed and closely spaced drops, the pairing instability is accompanied by a cascade of pair-switching events.

Our results show that the dynamics of laterally perturbed linear arrays are sensitive to particle-deformation effects. A transverse displacement of the trailing particle in an array of rigid spheres results in a decomposition of the array into a disordered group of particles. By contrast, arrays of deformable drops are more stable with respect to lateral displacements, and alignment with the flow tends to be restored during the evolution.

We have shown that pairwise quasi-two-dimensional hydrodynamic interactions qualitatively explain these phenomena. For arrays aligned in the flow direction, dipolar interparticle interactions cause the particles (drops or rigid spheres) to slow down. Since the trailing particle has only one neighbor, it moves faster than the other particles, thus it approaches the next particle in the array, and initiates the pairing instability. Dipolar interactions are also responsible for the instability of the trailing end of rigid-sphere arrays to lateral displacements.

Deformation endows drops with an additional quadrupolar far-field interaction which is absent for rigid spheres. The quadrupolar interaction draws drops together and tends to align them with the imposed velocity, thus stabilizing the array to lateral displacements. A balance between the deformation-induced far-field attraction and short-range hydrodynamic repulsion establishes a finite stable separation between the drops in a pair. By symmetry, quadrupolar interactions and hydrodynamic pairwise attraction/repulsion do not occur for rigid spheres.

We expect that the hydrodynamic mechanisms described in our paper, which are responsible for the collective dynamics of linear particle arrays, may also be relevant for explaining the collective dynamics of more complex confined dispersion flows. In particular, quadrupolar stabilization of flow-aligned drop arrays (*cf.* Fig. 19) is the likely mechanism underlying the spontaneous formation of droplet strings in confined emulsions under shear.^{51–55} Exploration of such phenomena will be described in forthcoming publications. The pairwise superposition approximation proposed and validated herein will be used to facilitate the computations.

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